

Practical #4

Practical Title:

To apply GitHub-based collaboration techniques including merging and remote repository management

Theory:

1. What is Merging in Git?

Merging in Git is the process of combining changes from one branch into another (usually into the main branch).

It helps integrate new features or bug fixes into the main project.

2. Types of Merge

◆ Fast-Forward Merge:

- Happens when there is no divergence
 - Branch moves forward directly
 - No extra merge commit created
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◆ Three-Way Merge:

- Happens when branches have different commits
 - Creates a **merge commit**
 - May cause conflicts
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3. What is Remote Repository?

A remote repository is a version of your project hosted online (like on GitHub).

It allows collaboration, backup, and sharing of code.

4. Common Remote Commands

- `git remote` → Shows connected remote repositories
- `git push` → Uploads local changes to remote
- `git pull` → Downloads + merges changes
- `git fetch` → Downloads changes without merging

Procedure:

Step 1: Create Local Repository

Step 2: Add Remote Repository

Step 3: Create Branch and Commit Changes

Step 4: Push Branch to Remote

Step 5: Merge Branch into Main

Step 6: Push Merged Code

Step 7: Fetch and Pull Updates

Output Screenshots:

```
Cloning into 'awesome-github-projects'...
remote: Enumerating objects: 762, done.
remote: Counting objects: 100% (50/50), done.
remote: Compressing objects: 100% (26/26), done.
remote: Total 762 (delta 45), reused 27 (delta 24), pack-reused 712 (from 3)
Receiving objects: 100% (762/762), 1.30 MiB | 242.00 KiB/s, done.
Resolving deltas: 100% (490/490), done.

$ git remote -v
origin https://github.com/mrudangwanjari/Agent_hub.git (fetch)
origin https://github.com/mrudangwanjari/Agent_hub.git (push)

$ git merge feature1
Already up to date.

$ git checkout feature1
Switched to branch 'feature1'

$ git log
commit 04096f1e0ba8d5990147d6a4d787e13c54bb6fd7 (HEAD -> main, origin/main, origin/HEAD)
Author: mrudangwanjari <wmrudang@gmail.com>
Date: Wed Feb 11 15:14:40 2026 +0530

    first commit

commit bebe430243726aaf79b11b089ee8a5b401eb81af
Merge: 4c7ecdb 0cfcf85
Author: mrudangwanjari <wmrudang@gmail.com>
Date: Tue Feb 10 11:43:05 2026 +0530

    Merge branch 'main' of https://github.com/Maheen-M02/Agent_Hub

commit 4c7ecdb80744fd6df443ac74a2594bb36880d8a
Author: mrudangwanjari <wmrudang@gmail.com>
Date: Tue Feb 10 11:42:40 2026 +0530

    changes through kiro

commit 0cfcf855cd7d35ee9f88616dc10a3f86adc47d79
Author: Maheen Meshram <maheenmeshram.skill102@gmail.com>
Date: Tue Feb 10 10:43:23 2026 +0530

    Delete .bolt directory

commit 6c36eb1dc0d3b6294fb432d76433010a8378446a
Merge: 3c96d6b eaf166d
Author: Maheencool <maheenmeshram.2115@gmail.com>
Date: Tue Feb 10 10:42:24 2026 +0530

    Merge branch 'main' of https://github.com/Maheen-M02/Agent_Hub

commit 3c96d6b88f5806761276a546f6d83e037e9893ce
Author: Maheencool <maheenmeshram.2115@gmail.com>
Date: Tue Feb 10 10:41:54 2026 +0530

    changes
```

```
$ git branch
* main

$ git pull
Already up to date.

$ git checkout
Your branch is up to date with 'origin/main'.
```

Activity / Task:

Task 1:

Create a remote repository and link it with local repository.

git init

git add .

git commit -m "Initial commit"

git remote add origin <https://github.com/sharvayuzade/DevOps-SEM-VI.git>

git push -u origin main

Repository Link: <https://github.com/sharvayuzade/DevOps-SEM-VI>

Task 2:

Perform branch merging and push changes to GitHub.

Commands Used:

```
git branch feature
git checkout feature
echo "New update" >> file.txt
git add .
git commit -m "Feature added"

git checkout main
git merge feature
git push origin main
```

Task 3:

Explain difference between git fetch and git pull.

Feature	git fetch	git pull
Action	Downloads changes	Downloads + merges
Merge	No	Yes
Safety	Safer (no auto merge)	Can cause conflicts
Usage	Review changes first	Direct update

Advantages

1. Easy remote collaboration
2. Backup of code online
3. Efficient branching and merging
4. Faster development workflow
5. Supports distributed teams

Challenges

1. Merge conflicts
2. Managing multiple branches
3. Requires proper understanding of commands